INF2009F Modelling using UML Exercise 2 --------13 Marks	
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Marks	/13

IMPORTANT NOTE:

- **Copied or Plagiarised answers will get 0 and student will be referred to Elsje Scott and Ayanda Pekane for further actions**
- **Please upload your completed exercise on Vula**

1. **Explain why models are useful during the requirements analysis phase of a software development lifecycle? (5 marks)**

Models are useful in the requirements phase because they can help break down complex problems into smaller, simpler problems and help identify other requirements needed. Models are also useful in providing clarity for developers in what the requirements fully entail and help the developers understand what the requirements are. Thus models allow people to understand what the clients require and thus produce a better product. Having models also allows for changes to be made to the requirements if there is a difference in the expectation between the client and company. The models are also useful in showing how the system will function, with a Use Case Diagram showing interactions between the user and the system. This is helpful for developers and makes the implementation of this quicker and easier. You can also see the structure of the system (class diagram), messages between the different objects in the system (sequence diagram) and activities (activity diagram). All this planning improves the development process and can safe resources (time and money).

2. **Why is it important to articulate the user requirements in a clear and succinct manner? (3 marks)**

Articulating user requirements clearly and well can lead to several advantages for a business. The most notable one is a better end result for the client. The client will be happier with the final product, with the description being better suited to what they want as a final product with the product working as expected. Another advantage is that time and resources can be reduced. A well-defined user requirement means time will not be wasted on requirements that the user does not need or on ambiguous requirements, saving money and time. A well-articulated user requirements will also help come up with the solution to the problem which can reduce time taken.

3. **What is the difference between functional and non-functional requirements (do some research)? (5 marks)**

The functional requirements are the basic requirements that must be present for the system to be functional. On the other hand, non-functional requirements state what the quality and performance of the system should be. The functional requirements are given by the customer and is therefore mandatory to be completed, whereas the non-functional requirements are specified by technical people in the company are therefore not compulsory to be met. Since the functional requirements are functions the user wants, they are easier to specify compared to the non-functional requirements. Another difference is that the functional requirements are given for each component of the system, whereas the non-functional requirements are a set of requirements that apply to the entire system

Examples for functional requirements are signing into a system or allowing a password to be reset whereas examples of non-functional requirements include allowing a certain number of users on the system at the same time and specified loading times in the system.